

# PRODUCT SPECIFICATION

Lithium Iron Phosphate  
LFP (LiFePO<sub>4</sub>) Battery

For (Battery Energy Storage System)

|         |                                    |
|---------|------------------------------------|
| Product | LFP (LiFePO <sub>4</sub> ) Battery |
| Model   | RP_LFP-314                         |
| Spec.   | 3.2V / 314Ah                       |
| Date    | 2025-12-10 ~ 2026-06-30            |

| Prepared | QA Checked | Approved |
|----------|------------|----------|
|          |            |          |
| 12 / 10  | 12 / 10    | 12 / 10  |

Remplir Inc.



## Revision History

| Ver. | Date       | Mark | Revision Description                    | Originator               |
|------|------------|------|-----------------------------------------|--------------------------|
| V1.0 | 2025-01-11 | /    | <b>First Edition</b>                    | <b>R&amp;D<br/>David</b> |
| V1.1 | 2025-04-20 | /    | Change format                           | QA/QC<br>Edward          |
| V2.0 | 2025-07-01 | /    | Add battery test conditions             | QA/QC<br>Edward          |
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|      |            |      |                                         |                          |
|      |            |      |                                         |                          |
|      |            |      |                                         |                          |

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## PART 1.

### Scope

- This specification is applied to the prismatic 314 Ah LFP battery cell of the square aluminum case supplied by Remplir Inc. Customer communicates with Remplir Inc. And if any special product specification is required, it should be agreed upon.

**Model Name : RP\_LFP-314**

## PART 2.

### Definition of Terms

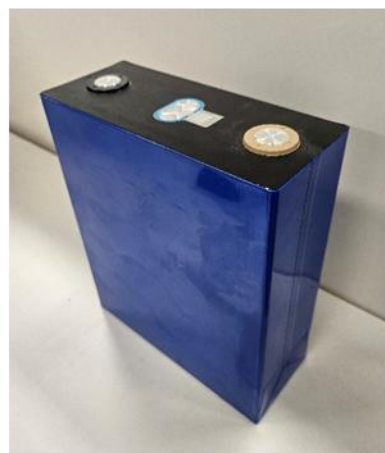
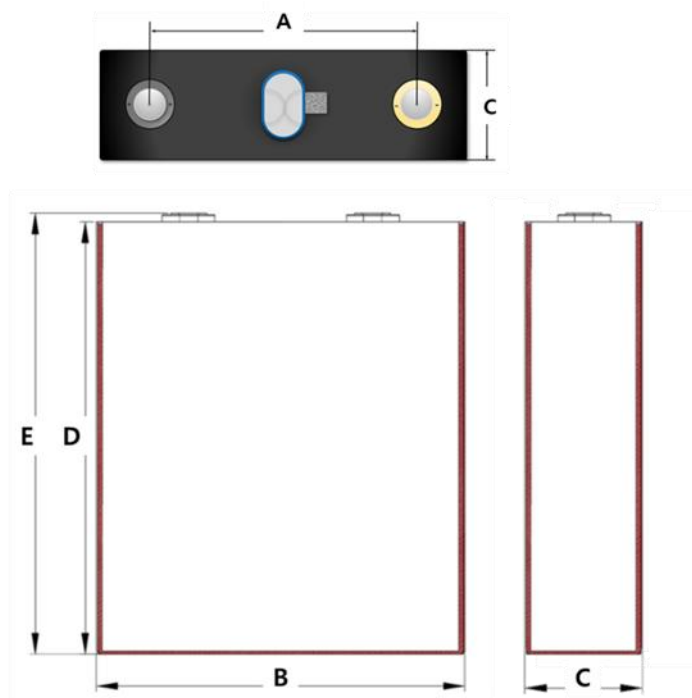
#### • Definition of Terms

| No   | Terms                      | Definition                                                                                                                                                                                                                                                                                                                 |
|------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.1  | Product                    | In this specification, "Product" refers to the prismatic-type <b>3.2V 314 Ah Lithium Iron Phosphate (LiFePO<sub>4</sub>) battery cell</b> manufactured by <b>REMPILIR INC.</b> (hereinafter, Remplir).                                                                                                                     |
| 2.2  | Ambient Temperature        | This refers to the temperature of the surrounding environment where the battery cells are operated or located.                                                                                                                                                                                                             |
| 2.3  | Cell Temperature           | Temperature measured by a temperature sensor installed at the center of the cell's surface. The selection of the temperature sensor and the measurement line is to be jointly decided by Remplir and the customer.                                                                                                         |
| 2.4  | C-Rate                     | The ratio of the charge/discharge current to the cell's capacity. For example, if the cell capacity is 314 Ah and the charge/discharge current is 157 A, the charge/discharge rate is 0.5 C. If the cell capacity decreases to 300 Ah and the charge/discharge current is 150 A, the charge/discharge rate is still 0.5 C. |
| 2.5  | State of Charge (SOC)      | The ratio of the cell's capacity to its nominal capacity under no-load conditions. It is measured in Ah or Wh and indicates the cell's charging status. For example, if 314 Ah cell has 314 Ah / 157 Ah / 0 Ah capacity, SOC will be 100 % / 50 % / 0 % SOC, respectively.                                                 |
| 2.6  | Cycle                      | The process of a single charging and discharging loop according to specified charge/discharge criteria.                                                                                                                                                                                                                    |
| 2.7  | State of Health (SOH)      | The ratio of the actual cell capacity to the nominal capacity after being used for a certain period. For example, a new cell with a nominal capacity of 314 Ah is considered to have 100 % SOH. If the cell capacity decreases to 251 Ah, it is considered to have 80 % SOH. A completely discarded cell has 0 % SOH.      |
| 2.8  | Open Circuit Voltage (OCV) | This is the cell voltage measured under no-load conditions or when the circuit is disconnected.                                                                                                                                                                                                                            |
| 2.9  | Depth of Discharge (DOD)   | This refers to the ratio of the cell's discharged capacity to its nominal capacity.                                                                                                                                                                                                                                        |
| 2.10 | Customer                   | This refers to the buyer signed in the sales contract with Remplir Inc.                                                                                                                                                                                                                                                    |

## PART 3.

## Product Dimensions

- Product Dimension (mm):



| List   | Description                   | Dimension      |
|--------|-------------------------------|----------------|
| A      | Distance between pole centers | 124.0 ± 0.3 mm |
| B      | Width                         | 173.4 ± 0.5 mm |
| C      | Thickness (100 ± 10 kgf)      | 71.7 ± 0.5 mm  |
|        | Thickness (Free)              | 71.7 ± 0.5 mm  |
| D      | Height (Shoulder)             | 204.2 ± 0.5 mm |
| E      | Height (with terminal)        | 206.8 ± 0.5 mm |
| Weight | Weight                        | 5.55 ± 0.15 kg |

\*The Battery pole is composed of dual-layered structure.

\*Laser welding is recommended for connecting and assembling corresponding module/pack.

\*The image shown is for reference only.

**PART 4.**
**Product Specification**
**● Summarized Product Specification**

| No.  | Items                               | Specification                                                                                                                                                              | Remark                                                                                                                                                         |
|------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.1  | Nominal Capacity                    | 314 Ah                                                                                                                                                                     | Standard Charging Method<br>Standard Discharging Method                                                                                                        |
| 4.2  | Nominal Energy                      | 1004.8 Wh                                                                                                                                                                  |                                                                                                                                                                |
| 4.3  | Nominal Voltage                     | 3.2 V                                                                                                                                                                      |                                                                                                                                                                |
| 4.4  | Operation Voltage                   | 2.5 — 3.65 V                                                                                                                                                               | Cell temperature<br>$0\text{ }^{\circ}\text{C} < T \leq 60\text{ }^{\circ}\text{C}$                                                                            |
|      |                                     | 2.0 — 3.65 V                                                                                                                                                               | $-20\text{ }^{\circ}\text{C} < T \leq 0\text{ }^{\circ}\text{C}$                                                                                               |
| 4.5  | Standard Charging Method            | CC(Constant current) charging at 0.5 C to 3.65 V, then CV(constant voltage) charging at 3.65 V until charging current declines to $\leq 0.05\text{ C}$ and rest for 30min. | At $(25 \pm 2)\text{ }^{\circ}\text{C}$<br>$0.5\text{ C} = 157\text{ A}$<br>$0.05\text{ C} = 15.7\text{ A}$<br>Averaged capacity after the continuous 3 cycles |
| 4.6  | Standard Discharging Method         | CC Discharging at 0.5 C to 2.5 V, and rest for 30min.                                                                                                                      |                                                                                                                                                                |
| 4.7  | AC-IR Cell Impedance                | $\leq (0.18 \pm 0.05)\text{ m}\Omega$                                                                                                                                      | Internal resistance is measured at 1.0 kHz of AC frequency at SOC 35 %                                                                                         |
| 4.8  | Maximum Continuous Charging Rate    | 0.5 C                                                                                                                                                                      | At $(25 \pm 2)\text{ }^{\circ}\text{C}$                                                                                                                        |
| 4.9  | Maximum Continuous Discharging Rate | 0.5 C                                                                                                                                                                      |                                                                                                                                                                |
| 4.10 | Cycle Life                          | $25\text{ }^{\circ}\text{C } 0.5\text{ C} / 0.5\text{ C}, \geq 6000\text{ cycles}$                                                                                         | 100 % DOD , 80 % SOH,                                                                                                                                          |
| 4.11 | Operation Temperature Range         | Charging: $0\text{ }^{\circ}\text{C} - 60\text{ }^{\circ}\text{C}$<br>Discharging: $-20\text{ }^{\circ}\text{C} - 60\text{ }^{\circ}\text{C}$                              | Cell temperature during the operation                                                                                                                          |
| 4.12 | Operation Humidity Range            | 10 — 95 % RH (Relative humidity)                                                                                                                                           | Without precipitation                                                                                                                                          |
| 4.13 | Shipment Capacity                   | $(35 \pm 5)\text{ \% SOC}$                                                                                                                                                 | Within 15 days from factory                                                                                                                                    |
| 4.14 | Structural Stability                | No stain, No rupture,<br>No deformation, No leakage                                                                                                                        | -                                                                                                                                                              |
| 4.15 | Standard Testing Condition          | Temperature: $(25 \pm 2)\text{ }^{\circ}\text{C}$ / Humidity: $\leq 80\text{ \% RH}$<br>Atmospheric pressure: 86 — 106 kPa                                                 |                                                                                                                                                                |

**PART 4.**
**Product Specification**
**• Detailed Product Specification**

| No.  | Items                                  | Specification                                                                           | Remark                                                                                                                                                      |
|------|----------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.16 | Maximum Pulse Charging Rate            | 1.0 C (50 % SOC, 30 sec)                                                                | At (25 ± 2) °C                                                                                                                                              |
| 4.17 | Maximum Pulse Discharging Rate         | 1.0 C (50 % SOC, 30 sec)                                                                |                                                                                                                                                             |
| 4.18 | Discharge Capacity at High Temperature | ≥ 98 %                                                                                  | At (60 ± 2) °C , 1 C , 2.5 V                                                                                                                                |
| 4.19 | Discharge Capacity at Low Temperature  | ≥ 70 %                                                                                  | At (-20 ± 2) °C , 1 C , 2.0 V                                                                                                                               |
| 4.20 | Charging Temperature Range             | 0 — 60 °C                                                                               | Regardless of the battery's SOC level or charging mode, charging stops if the cell temperature exceeds the absolute charging temperature range.             |
| 4.21 | Discharging Temperature Range          | -20 — 60 °C                                                                             | Regardless of the battery's SOC level or discharging mode, discharging stops if the cell temperature exceeds the absolute discharging temperature range.    |
| 4.22 | Long-term Storage Temperature Range    | -20 — 45 °C ≤ One month<br>(20 — 50 % SOC)<br>0 — 35 °C ≤ Six months<br>(25 — 50 % SOC) | For storage periods exceeding 6 months, it is recommended to inspect the battery condition before use.<br>Storage humidity ≤ 90 % RH without precipitation. |
| 4.23 | Self-discharging Rate                  | ≤ 3.0 % per month                                                                       | At (25 ± 2) °C<br>(30 — 50 % SOC)                                                                                                                           |
| 4.24 | Welding Parameters for Pole & Bus-bar  | Max welding depth ≤ 1.7 mm                                                              | No leakage                                                                                                                                                  |
|      |                                        | Max pressure ≤ 1000 N                                                                   | No deformation                                                                                                                                              |
|      |                                        | Max torque ≤ 5 Nm                                                                       | No loosening                                                                                                                                                |
|      |                                        | Max temperature ≤ 170 °C                                                                | 1 sec of welding, No deformation                                                                                                                            |

**PART 5.**
**Charging and Discharging Conditions**
**5.1 Charging mode**

| No.   | Items                                      | Specification                                                                                                             | Condition                                                                                                                      |
|-------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 5.1.1 | Standard Charging Rate (Current)           | 0.5 C                                                                                                                     | At (25 ± 2) °C                                                                                                                 |
| 5.1.2 | Maximum Charging Rate (Current)            | 0.5 C                                                                                                                     |                                                                                                                                |
| 5.1.3 | Standard Charging Voltage                  | ≤ 3.65 V (single cell)                                                                                                    |                                                                                                                                |
| 5.1.4 | Standard Charging Method                   | CC charging at 0.5C to 3.65 V, then CV charging at 3.65 V until charging current declines to ≤0.05 C and rest for 30min.. |                                                                                                                                |
| 5.1.5 | Standard Charging Temperature              | At (25 ± 2) °C                                                                                                            |                                                                                                                                |
| 5.1.6 | Absolute Charging Temperature Range (Cell) | 0 — 60 °C                                                                                                                 | Regardless of the battery's SOC level, charging stops if the cell temperature exceeds the absolute charging temperature range. |
| 5.1.7 | Absolute Charging Voltage (Cell)           | 3.65 V                                                                                                                    | Regardless of the battery's SOC level, charging stops if the cell voltage exceeds the absolute charging voltage.               |

**5.2 Temperature-based charging conditions (C-Rate)**

| Cell temperature / ( °C )            |               | < 0         | 10    | 20    | 25    | 45    | 55    | > 60        |
|--------------------------------------|---------------|-------------|-------|-------|-------|-------|-------|-------------|
| Maximum continuous Charging Rate (C) | 0 — 100 % SOC | Not allowed | 0.5 C | 0.5 C | 0.5 C | 0.5 C | 0.5 C | Not allowed |

**5.3 Discharging mode**

| No.   | Items                                       | Specifications                                                                                                                                                     | Condition                                                                                                                                       |
|-------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.3.1 | Standard Discharge Rate (Current)           | 0.5 C                                                                                                                                                              | At (25 ± 2) °C                                                                                                                                  |
| 5.3.2 | Maximum Discharge Rate (Current)            | 0.5 C                                                                                                                                                              |                                                                                                                                                 |
| 5.3.3 | Standard Discharge Voltage                  | 2.5 V                                                                                                                                                              | T > 0 °C                                                                                                                                        |
|       |                                             | 2.0 V                                                                                                                                                              | T ≤ 0 °C                                                                                                                                        |
| 5.3.4 | Standard Discharge Method                   | When the voltage reaches 2.5 V after performing a 0.5 C constant current discharge, the discharge is terminated and the battery is allowed to rest for 30 minutes. |                                                                                                                                                 |
| 5.3.5 | Standard Discharge Temperature              | At (25 ± 2) °C                                                                                                                                                     |                                                                                                                                                 |
| 5.3.6 | Absolute Discharge Temperature Range (Cell) | -20 — 60 °C                                                                                                                                                        | Regardless of the discharge mode the battery is in, discharging stops if the cell temperature exceeds the absolute discharge temperature range. |

|       |                                   |       |                                                                                                                                           |  |  |  |  |
|-------|-----------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|
| 5.3.7 | Absolute Discharge Voltage (Cell) | 2.0 V | Regardless of the discharge mode the battery is in, discharging stops when the cell voltage exceeds the absolute discharge voltage range. |  |  |  |  |
|-------|-----------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|

5.4 Temperature-based discharging conditions (C-Rate)

| Cell temperature / ( °C )               |               | < -20       | 0     | 15    | 25    | 45    | 55    | > 60        |
|-----------------------------------------|---------------|-------------|-------|-------|-------|-------|-------|-------------|
| Maximum continuous Discharging Rate (C) | 0 — 100 % SOC | Not allowed | 0.5 C | 0.5 C | 0.5 C | 0.5 C | 0.5 C | Not allowed |

\*Actual battery charging/discharging Methods may be varied depending on the usage conditions agreed between the manufacturer (Remplir Inc.) and the customer.

## PART 6.

## Battery Performance Test

### 6.1 Visual Inspection

- The battery must be free from any defects such as scratches, cracks, rust, stains, or electrolyte leakage that may adversely affect its performance.
- During battery performance testing, a new battery shall be used for each test and test condition to ensure accurate and independent evaluation of performance.

### 6.2 Environmental Test Conditions

- Unless otherwise specified, all tests described in this product specification shall be conducted under the following test conditions:
- Temperature: (25 ± 2) °C, Relative Humidity (RH): ≤ 80 %, Pressure: 86 kPa – 106 kPa

### 6.3 Test Equipment Accuracy

|                         |                       |                  |                   |
|-------------------------|-----------------------|------------------|-------------------|
| Voltage measurement     | Accuracy: ≥ 0.5 grade | Size measurement | Accuracy: ± 0.1 % |
| Current measurement     | Accuracy: ≥ 0.5 grade | Time measurement | Accuracy: ± 0.1 % |
| Temperature measurement | Accuracy: ± 0.5 °C    |                  |                   |

### 6.4 Performance Test methods

| No.   | Items             | Test methods and conditions                                                                                                                                                                                                                                                                                                  | Standards                     |
|-------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 6.4.1 | Capacity & Energy | 1) CC discharging the cell to 2.5 V at 0.5 C, then rest for 30 minutes.<br>2) Charging the cell according to the standard charging method (5.1.4).<br>3) Discharging the cell according to the standard discharging method (5.3.4).<br>4) Repeating steps 2) through 3) three times, and measuring the third capacity as C0. | C0 ≥ 314 Ah<br>C0 ≥ 1004.8 Wh |

|       |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                        |
|-------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.4.2 | Rated Capacity              | <p>1) Charging the cell according to the standard charging method (5.1.4).</p> <p>2) CC discharging the cell to 2.5 V at 0.5 C or 1 C, respectively, and recording the discharge capacity as C1.</p> <p>3) Calculating C1/nominal capacity as the capacity retention ratio.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p><math>\geq 100\%</math> at 0.5 C</p> <p><math>\geq 98\%</math> at 1.0 C</p>                                                                                         |
| 6.4.3 | Temperature Characteristics | <p><b>(A) High-Temperature Characteristics:</b></p> <p>1) Charging the cell according to the standard charging method (5.1.4).</p> <p>2) Storing the cell at <math>(55 \pm 2)^\circ\text{C}</math> for 5 hours, then CC discharging it to 2.5 V at 1 C, and recording the discharge capacity as C2.</p> <p>3) Calculating C2/nominal capacity as the capacity retention ratio.</p> <p><b>(B) Low-Temperature Characteristics:</b></p> <p>1) Charging the cell according to the standard charging method (5.1.4).</p> <p>2) Storing the cell at <math>(-20 \pm 2)^\circ\text{C}</math> for 24 hours, then CC discharging it to 2.0 V at 1 C, and recording the discharge capacity as C2.</p> <p>3) Calculating C2/nominal capacity as the capacity retention ratio.</p> | <p>At <math>55^\circ\text{C}</math>: Capacity Retention <math>\geq 95\%</math></p> <p>At <math>-20^\circ\text{C}</math>: Capacity Retention <math>\geq 70\%</math></p> |
| 6.4.4 | Cycle Life                  | <p>1) Charging the cell according to the standard charging method (5.1.4).</p> <p>2) Discharging the cell according to the standard discharging method (5.3.4).</p> <p>3) Repeating the above steps until the discharge capacity drops below 80 % of the cell's nominal capacity.</p> <p>*The cell is preloaded with 20 to 120 kgf at 50 % SOC.</p>                                                                                                                                                                                                                                                                                                                                                                                                                    | <p><math>\geq 80\%</math> of Nominal Capacity</p>                                                                                                                      |
| 6.4.5 | Storage Characteristics     | <p>1) Charging the cell according to the standard charging method (5.1.4).</p> <p>2) Storing the cell for 28 days without an external load circuit.</p> <p>3) Discharging the cell according to the standard discharging method (5.3.4) and recording the discharge capacity as the residual capacity.</p> <p>4) Again, Charging the cell according to the standard charging method (5.1.4), discharging it according to the standard discharging method (5.3.4), and recording the discharge capacity as the recovered capacity.</p>                                                                                                                                                                                                                                  | <p>Residual Capacity <math>\geq 95\%</math> of Nominal Capacity</p> <p>Recoverable Capacity <math>\geq \text{Nominal Capacity} \times 97\%</math></p>                  |

**PART 7.**
**Battery Safety Test**
**• List of Battery Safety Test**

| No. | Items                     | Test methods and Conditions                                                                                                                                                                                                                                                                                                                                                                                                           | Requirements                            |
|-----|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 7.1 | Over-charging Test        | 1) Fully charging of the cell based on the charging method (5.1.4)<br>2) Charging the battery to 1.5 times higher voltage than maximum voltage OR CC charging at 1 C for 1 hour<br>3) Observing the battery condition for 1 hour                                                                                                                                                                                                      | No explosion,<br>No fire                |
| 7.2 | Over-discharging Test     | 1) Fully charging of the cell based on the charging method (5.1.4)<br>2) CC discharging at 1 C for 90 min. OR until the voltage reaches 0 V<br>3) Observing the battery condition for 1 hour                                                                                                                                                                                                                                          | No explosion,<br>No fire                |
| 7.3 | Short-circuit Test        | 1) Fully charging of the cell based on the charging method (5.1.4)<br>2) Directly connecting the positive and negative terminals of the battery with a wire (with a resistance of $\leq 5 \text{ m}\Omega$ ) for 10 minutes<br>3) Observing the battery condition for 1 hour                                                                                                                                                          | No explosion,<br>No fire                |
| 7.4 | Pressure Test (Extrusion) | 1) Fully charging of the cell based on the charging method (5.1.4)<br>2) By using a semicircular crimping tool with a radius of 75 mm, pressing the battery body vertically at a speed of $(5 \pm 1) \text{ mm/s}$ .<br>- Pressing the battery until the voltage reaches 0 V, OR, the structural deformation ratio reaches 30 %, or, the pressure reaches $(13 \pm 0.78) \text{ kN}$<br>3) Observing the battery condition for 1 hour | No explosion,<br>No fire                |
| 7.5 | Drop Test                 | 1) Fully charging of the cell based on the charging method (5.1.4)<br>2) Free-dropping the battery from a height of 1.5 M onto a solid flat surface with the battery terminals facing down<br>3) Observing the battery condition for 1 hour                                                                                                                                                                                           | No explosion,<br>No fire                |
| 7.6 | Low Pressure Test         | 1) Fully charging of the cell based on the charging method (5.1.4)<br>2) Storing the battery in a low-pressure environment (11.6 kPa) for 6 hours<br>3) Observing the battery condition for 1 hour                                                                                                                                                                                                                                    | No explosion,<br>No fire,<br>No leakage |
| 7.7 | Heating Test              | 1) Fully charging of the cell based on the charging method (5.1.4)<br>2) Placing the battery in an oven and increasing the temperature to $(130 \pm 2) ^\circ\text{C}$ at a rate of $5 ^\circ\text{C/min}$ , then hold for 30 minutes<br>3) Observing the battery condition for 1 hour                                                                                                                                                | No explosion,<br>No fire                |

**\*Test environment:** Temperature at  $(25 \pm 2) ^\circ\text{C}$  / Humidity:  $\leq 80 \text{ \% RH}$  / Pressure: 86 — 106 kPa

## PART 8.

## Safety Instructions

### 8.1 Safety Instructions for Battery Utilization

- Battery cells contain flammable materials, such as organic solvents. Improper handling may result in fire, smoke generation, or explosion, and can cause severe damage to the battery's functionality.
- When integrating the battery into a module or pack, it is essential to design and implement appropriate safety and protection systems, including a Battery Management System (BMS) and protective circuits, to ensure safe operation and prevent hazardous conditions.
- Subsequent manufacturers are strongly recommended to include these safety guidelines in their user manuals to ensure proper handling and end-user safety.
- Please read and understand the prohibited actions carefully before using or assembling the battery.

### 8.2 Cell Swelling and Preload Requirements

- Preload force must be applied during safety testing, cycle life testing, and pack design to ensure mechanical stability and safety.
- The recommended clamping force range for new cells is 2,000 – 5,000 N, with an allowable tolerance of  $\pm 200$  N.
- Swelling (expansion) of cells may be occurred during the long-term operation. The measured expansion force under the specified test conditions (At 70 % SOH, 15 mm steel plate) is approximately 25,000 N.
- Customers should consider the reliability of structural strength during the product design process.

### 8.3 Cell Protection Functions:

- Cells must have over-charge protection, over-discharge protection, and over-current protection during use. The functional requirements for the BMS and protection circuit are as follows:

| No.   | Items                               | Parameter condition                                       | Protective action                                                                                 |
|-------|-------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 8.3.1 | Primary over-charge protection      | $\geq 3.70$ V                                             | Setting the charging current limit to 0 A                                                         |
| 8.3.2 | Secondary over-charge protection    | $\geq 3.80$ V                                             | Forced termination of charging                                                                    |
| 8.3.3 | Primary over-discharge protection   | $\leq 2.4$ V ( $> 0$ °C )<br>$\leq 1.9$ V ( $\leq 0$ °C ) | Reducing the current to the minimum allowable value                                               |
| 8.3.4 | Secondary over-discharge protection | $\leq 2.2$ V ( $> 0$ °C )<br>$\leq 1.7$ V ( $\leq 0$ °C ) | Forced termination of discharging                                                                 |
| 8.3.5 | Over-heating protection             | Battery temperature<br>$\geq 60$ °C                       | Termination of charging/discharging when the temperature of the cell reaches the limit            |
| 8.3.6 | Over-current protection             | Charge/discharging current<br>> Max. continuous current   | Termination of charging/discharging when the current reaches the maximum continuous limit current |
| 8.3.7 | Short-circuit protection            | -                                                         | Protection of terminals and electrically conductive parts                                         |

**8.4 Prohibitions:**

- Do not immerse the battery cell in water, beverages, or any other liquids.
- Do not incinerate or heat the battery cell. Avoid using or storing it near heat sources or in high-temperature environments.
- Do not reverse the positive/negative electrode polarity or install the electrodes in the wrong direction.
- Do not short-circuit the positive and negative terminals by connecting them directly with metal objects (e.g., wires, metallic tools).
- Do not apply excessive impact such as hitting, throwing, or stepping on the battery cell.
- Do not pierce, drill or disassemble the battery cell with nails or other sharp objects.

**8.5 Warning:**

- Do not place the battery in or near a microwave oven or any other cooking appliance. Exposure to high temperatures or strong electromagnetic radiation may cause the battery leakage, heat or smoke generation, fire, or explosion.
- Do not combine or connect Remplir's batteries with other manufacturer's batteries with different capacities or chemistries. Using mismatched batteries may result in overheating, smoke, fire or explosion.
- If any abnormal signs such as odor, heat, discoloration, or deformation are detected, immediately remove the battery from the device or charger and stop using it. Continued use of a defective battery may cause overheating, smoke, fire or explosion.
- If charging is not completed within the specified time, stop charging immediately.
- Do not use a leaking battery near open flames.
- Do not directly touch the leaking battery. If the leaked liquid comes into contact with your eyes, immediately wash it thoroughly with clean water and seek medical attention quickly. Failure to do so may result in serious eye injury.

**8.6 Note:**

- Do not use or store at high-temperatures environments. Excessive heat may cause overheating, ignition, or fire and lead to performance degradation and reduced battery lifespan.
- Charge the battery only at temperatures between 0 °C and 60 °C. Charging outside this range may result in electrolyte leakage, heat generation, reduced battery performance and lifespan.
- Contact your supplier immediately if you notice any signs of unusual odor, heat generation, or rust/corrosion before using the battery for the first time.

**PART 9.****Battery Cell / Module / Pack Assembly Precautions****9.1 Basic Precautions**

- Avoid contact with impact, high temperatures, or sharp objects during battery cell, module, or pack assembly.
- Always wear insulating gloves and protective equipment during assembly operations.
- Consult with the supplier regarding the design, assembly, and manufacturing process prior to implementation.

**9.2 Battery Cell Connection**

- Direct soldering of wires or devices to battery cells is strictly prohibited.
- Direct soldering can damage components such as separators and insulators due to heat generation.
- Apply appropriate cell connection and fastening methods according to the battery terminal shape and condition.
- Use bus bars of proper size, shape, and material designed to accommodate the required output characteristics. Laser welding is recommended for cell interconnection.
- Do not exceed the permitted torque value when tightening bolts. Otherwise, it can cause damages to terminals or connecting parts.

**9.3 Prevention of Internal Short-Circuits in Battery Cell/Module/Pack**

- For additional safety, insulation must be designed or applied between cells and to the module/pack body (frame) as required by the performance requirements. The battery module/pack must be designed to prevent internal short-circuits through internal structural design and insulation measures.

**9.4 Prohibition of Disassembly**

- Do not disassemble cells, modules, or packs without authorized companies and qualified personnel. Unauthorized disassembly may cause internal short-circuits, leading to gas generation, fire, explosion, and personal injury or equipment damage.
- If disassembly is required, contact the manufacturer or authorized personnel for proper handling.

**9.5 Prohibition of Improper Battery Cell/Module/Pack Disposal**

- Do not incinerate or dispose of battery cells/modules/packs in fire. Doing so is extremely dangerous and may cause the cells to explode.
- Do not spill or immerse battery cells/modules/packs in water, seawater, or any other liquids such as soft drinks, juice, or coffee. Exposure to liquids may cause malfunction, corrosion, short-circuit, or fire.

**9.6 Battery Cell/Module/Pack Replacement**

- Battery replacement shall only be performed by the cell supplier or module/pack/equipment manufacturer. Users must not attempt to replace the battery directly.
- Unauthorized replacement may result in equipment malfunction, safety hazards, or loss of warranty coverage.

**9.7 Prohibition of Damaged Battery Cell/Module/Pack Utilization**

- Cells may be damaged during transportation due to impact or mishandling. Do not use if the cell exhibits any abnormalities such as damage to the external film packaging, deformation of the cell packaging, electrolyte odor, or electrolyte leakage.
- Do not place cells/modules/packs that show signs of electrolyte odor or leakage near open flames or heat sources.

## PART 10.

### Warranty

#### 10.1 Warranty (General Cases)

- The warranty period for the battery is **1 years** from the manufacturer's shipping date. The manufacturer (Remplir Inc.) guarantees replacement of the battery if a defect is found to be caused by a manufacturing process issue, and not by customer negligence, misuse, or improper handling.

#### 10.2 Warranty (Special Cases)

- The warranty shall remain valid until the cell cycle count reaches the specified cycle life defined in this specification under standard conditions. In the event that a separate warranty contract exists, the terms of that contract shall take precedence.

## PART 11.

### Storage Method

#### 11.1 Storage Method (Condition)

- If the battery is to be stored for more than **1 month**, it shall be kept in a clean, dry environment with a relative humidity (RH) of 90 % or less and an ambient temperature between 10 °C and 35 °C. The battery must be kept away from corrosive substances, open flames, and heat sources. A charge and discharge cycle shall be performed once **every 6 months** during storage to maintain battery performance. At the time of shipment to the customer, the SOC of the battery shall be  $(35 \pm 5) \%$ .

## PART 12.

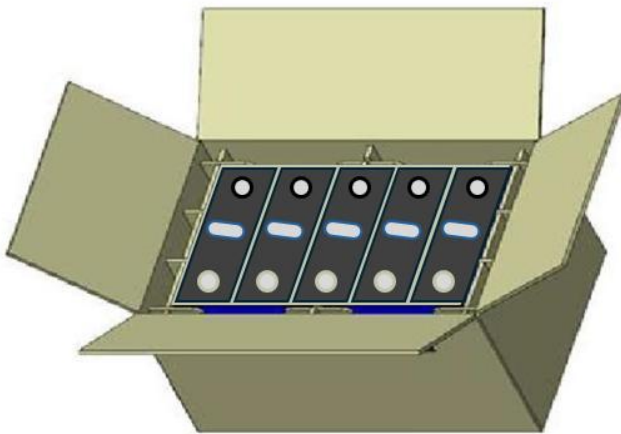
### Packaging and Transportation

#### 12.1 Product Packaging

- The batteries shall be shipped in a state of charge (SOC)  $(35 \pm 5) \%$ , and shall be transported in packaging designed to protect against mechanical shock and impact.
  - **For small-lot shipments and standard box packaging**, each corrugated cardboard box shall contain 6 cells, individually separated and packed with the terminal side facing upward.
  - **For bulk shipments using palletized packaging**, a standard configuration of 9 boxes per layer and 4 layers per pallet shall be applied. The entire pallet shall be securely wrapped to prevent the ingress of dust and moisture prior to shipment.
- Packaging units and conditions may be modified based on customer requirements and mutual agreement between the manufacturer (Remplir Inc.) and the customer.

#### 12.2 Product Transportation

- During transportation, the batteries shall be protected from excessive vibration, shock, and compression, and shall be kept away from direct sunlight and rain. It is recommended to use vehicles or containers that provide adequate protection against external contamination and mechanical impact. Depending on transportation conditions and the applicable certification scope, the batteries may be transported by road, rail, sea, or air.



L X W X H : 440 x 250 x 258 mm



L X W X H : 1100 x 1100 x 1150 mm

Images of battery cell BOX packaging (left) and pallet loading packaging (right)

## PART 13.

## Battery Handling and Lifespan Management

### 13.1 Handling of Batteries

- Since the battery operates based on chemical reactions, prolonged storage without use may result in gradual performance degradation over time. If operating conditions such as charging, discharging, and ambient temperature are not maintained within the specified limits, the battery life may be shortened, and electrolyte leakage may occur, potentially causing damage to the battery or the associated equipment. Even if the battery is charged under normal conditions, it may require replacement if it fails to maintain its charge after extended storage.

### 13.2 Battery Lifespan Management

- All the batteries have a limited lifespan. During operation, the customer shall establish an effective tracking system capable of monitoring and recording the battery's lifetime, as well as the internal resistance and capacity of each cell. The methods for measuring and calculating internal resistance and capacity shall be discussed and agreed upon between the manufacturer (Remplir Inc.) and the customer. If the internal resistance of a battery exceeds 150 % of its initial value, or if its capacity decreases below 70 % of the nominal capacity, the battery shall not be used. Failure to comply with these requirements shall release the manufacturer from any product quality warranty obligations under the sales contract and this specification.

**PART 14.****Agreements**

**Any other matters not included in this specification must be agreed between the manufacturer (Remplir Inc.) and the customer.**

**14.1** Any applications or operating conditions that differ from those specified in this document **must be discussed and approved** in advance by the manufacturer (Remplir Inc.). Additional testing may be required to verify the performance and safety of the product under such conditions.

**14.2** The manufacturer shall not be held responsible for any accidents, damages, or malfunctions resulting from the use of the battery under conditions **not specified in, or not approved in writing by**, this document.

**14.3** Any matters not covered in this specification shall be determined **through mutual agreement** between the manufacturer and the customer.

**14.4** This product complies with the <Eco-Friendly and Hazardous Substance Management Standards for Manufactured Products>.

**14.5** Manufacturer **reserves the right** to revise product specifications. The customer **must verify the latest version** of the product specification prior to purchase.

**14.6** In the event of any discrepancy or inconsistency between the English and Korean versions of this document, **the Korean version shall prevail.**

**PART 15.****Manufacturer Information**

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